Group 90 Project Proposal

Sleep Debt & Screen Time: Late Night Phone Habits

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Task Overview:

Smartphones are an integral part of our lives due to benefits; such as easy accessibility to information, social connectivity, workplace/financial applications, and entertainment including watching videos or playing video games. Smartphones have been revolutionized the ease of information and has everyone connected. However, the smartphone - with many pros stated - has also many drawbacks that has disconnected everyone from reality. Spending too much time on the screen reduced personal socialization and significant sleep [1]. Sleep plays a crucial role in developing new memory and is essential for all stages of learning. With inadequate amounts of sleep, the body and mind experience rapid physical and cognitive decline, affecting overall long-term health. These include mood swings, [2]. Over 3/4 of Americans lose sleep because of digital distractions [3]. The number might be different for different nations, but this is a severe problem that affects a lot of people.

The goal of this project is to analyze and predict the fatigue that people feel the following day based on their screentime usage.

There are a few challenges with trying to predict 'sleep fatigue' based on screentime. There are a lot of variables involved, they might not scale linearly with sleep loss, and there is a limit to the amount of sleep fatigue a person might experience (it could be different for different people).

Task Definition:

The data type is tabular containing number and category features like screen time, type of screen use, device type, and the resulting sleep patterns during weekdays and weekends. The task type is regression since we are trying to figure out the exact amount of sleep loss in hours. The number of classes is not applicable for this task since we are using regression. Since we are predicting sleep loss for both weekdays and weekends, this is a multi-label problem where we will get 2 outputs per set of inputs

Dataset: https://data.mendeley.com/datasets/pj9hzmp7ym/1

Data Collection:

Data will be downloaded directly via the Mendeley data repository. This data has a open-access creative commons terms license which means we are allowed to use it for our purposes. Metadata is listed below.

Metadata:

• Principal Investigators / Contributors: Santiago Perez-Lloret and Daniel Perez Chada

• Repository Platform: Mendeley Data

• Publication Date: October 5, 2022

Information about the dataset:

• Target Population: (1,257) school-aged teens aged 12 to 18 years across 52 urban and suburban schools in Argentina.

• Collection Mechanism: Self-administered structured surveys looking at routine habits, technology usage, sleep times and patterns, and standardized school performance (we will not look at school performance).

We will only use a small subset of the data because only part of it is relevant to predicting sleep scores. We won't look at school performance, subjective responses on how tired a student is feeling, or any other attributes that are not relevant to sleep/screentime or are not easily interpreted. This is because we don't want to crowd our model with a bunch of parameters that might not even affect our predicted result.

Proposed Solution:

We think we might end up using polynomial regression to train this model since we expect sleep loss to scale with hours of screentime before bed, but we know that after a certain point, there is a limit. There are already models that are used to predict different sleep metrics based on screentime such as the health app that is present on Samsung phones or the Sleep-Health-Digital-Screen-Exposure-Risk-Prediction repository on GitHub [4]. We will know if the model is performing well if we see that it predicts sleep loss accurately. For example, if the actual sleep loss is 3 hours but the model predicts 2 hours. The libraries that we intend to use are numpy, matplotlib, and sci-kit learn.

References [IEEE format]:

- [1] “Kids and young adults are addicted to screens. New Research argues it may be a public health concern | CBC News,” CBCnews, https://www.cbc.ca/news/health/youth-screen-time-tech-companies-9.7365434 (accessed Oct. 8, 2026).

- [2] A. S. Shah et al., “Effects of sleep deprivation on physical and Mental Health outcomes: An umbrella review,” American journal of lifestyle medicine, https://pmc.ncbi.nlm.nih.gov/articles/PMC12116485/ (accessed Oct. 8, 2026).

- [3] “Over three-fourths of Americans lose sleep due to digital distractions – sleep experts urge a change,” Newswise, https://www.newswise.com/articles/over-three-fourths-of-americans-lose-sleep-due-to-digital-distractions-sleep-experts-urge-a-change?ta=home (accessed Oct. 8, 2026).

- [4] “SOM-12211285/sleep-health-digital-screen-exposure-risk-prediction: A comprehensive data analysis and machine learning project exploring how digital screen exposure impacts sleep health. the project includes EDA, visualizations, and predictive modeling to uncover patterns and forecast sleep quality outcomes based on screen usage.,” GitHub, https://github.com/som-12211285/Sleep-Health-Digital-Screen-Exposure-Risk-Prediction (accessed Oct. 8, 2026).

Team Contract:

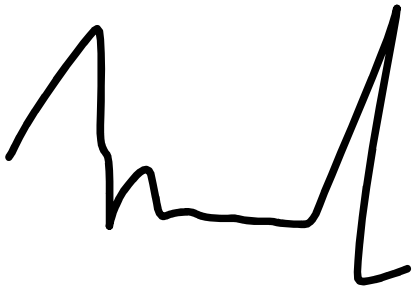
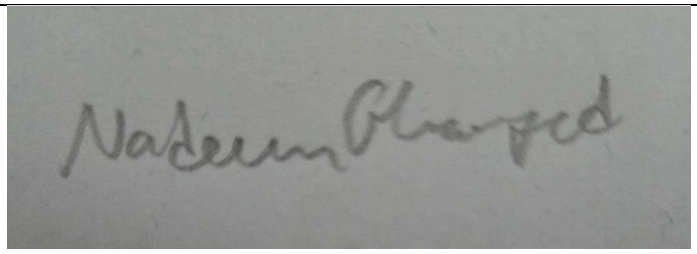
Team's purpose

The goal of the project is to be able to identify which factor has the biggest effect on the response variable

Duties

Handling

Signature

Name	Signature	Date
Muhammad Ullah		October 6, 2026
Nadeem Elsayed		October 6, 2026